

Augusto Salazar Montes

Software Acceptance Test Engineer | Embedded Validation Specialist

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Professional Profile

Software Acceptance Test Engineer with 4+ years of experience in embedded software validation, test automation, and system integration for the automotive industry. Proven track record in designing and executing black-box and white-box test campaigns, performing root cause analysis, and ensuring compliance with quality standards. Proficient in **Python** for test automation, **CAN/CAN-FD** protocols using **CANoe** and **CANalyzer**, and HIL/bench setups. Bilingual (English B2, French B1) with international research experience. Seeking to apply strong analytical and debugging skills in ADAS/automotive validation projects.

Technical Skills

Test Automation & Validation

Python (pandas, matplotlib, pytest), LabVIEW (Core 1&2), TestStand, CAPL (learning), HIL/bench setups, black-box/white-box testing, integration testing, regression testing

Automotive Protocols & Tools

CAN/CAN-FD, CANoe, CANalyzer, UART, I2C, SPI, LIN (basic), UDS Diagnostics (ISO 14229) – basic, Linux (basic)

Debugging & Root Cause Analysis

Oscilloscopes, logic analyzers, multimeters, log analysis, TRACE32 (awareness), iSystem (awareness)

Programming & Tools

Python, C, C++, VHDL, SQL, Git, GitHub Actions, Docker, MATLAB, Jira (familiar)

Languages

Spanish (Native), English (B2 Proficient – EF SET), French (B1 – TCF)

Work Experience

Application Engineer / Oct 2022 – Jan 2026

Seica, Inc. – Querétaro, MX

- Designed and executed **automated test campaigns** (functional, integration, regression) for automotive electronic modules using **LabVIEW** and **TestStand**, ensuring compliance with customer specifications.
- Used **CANoe** and **CANalyzer** to monitor, analyze, and simulate **CAN/CAN-FD communication** between ECUs and test fixtures, ensuring proper data exchange and signal integrity.
- Developed **Python scripts** to automate test log processing, analyze system performance, and generate KPIs (Test Yield, Defect Density, First Pass Yield) for stakeholders.
- Performed **root cause analysis** on hardware/software failures using oscilloscopes, logic analyzers, and multimeters, reducing defect escape rate.
- Managed **SQL databases** and **Git repositories** for end-to-end traceability, version control, and documentation of test protocols.
- Collaborated with **international cross-functional teams** (Mexico, US, Italy) to align test requirements and resolve production issues.
- Trained customer engineers (Marelli) and internal teams on test software operation, debugging, and maintenance.

Python LabVIEW TestStand CANoe CANalyzer CAN SQL Git Root Cause Analysis

- Implemented **data acquisition and process control systems** for automotive manufacturing clients, reducing manual validation cycle time.
- Created **LabVIEW-based automation frameworks** for real-time monitoring and control, integrating SQL for data logging and analysis.

LabVIEW SQL Data Acquisition Automation

- Programmed industrial control interfaces using **LabVIEW** and **TestStand** for real-time monitoring in manufacturing environments.
- Developed test plans, maintained **SQL databases**, and provided hardware/software troubleshooting support.

LabVIEW TestStand SQL HMI Industrial Control

- Designed and implemented **FPGA-based hardware solutions (VHDL)** for a particle detection system, processing high-frequency oscillator data in real-time.
- Gained hands-on experience with **low-level debugging**, timing analysis, and hardware-software integration.

VHDL FPGA Hardware Prototyping Xilinx

Key Projects

Automated KPI Reporting & Test Data Analysis (Python)

- Developed Python scripts using **pandas**, **matplotlib**, and **Tkinter** to automatically extract, process, and visualize test logs from industrial systems, generating KPIs (Test Yield, Defect Density, First Pass Yield) as PDF reports.

Automotive Injector Testing & Cleaning System (STM32)

- Designed full system: control (STM32), isolation (optocouplers 4N45), power (MOSFETs), custom PCBs, firmware in C, flyback diodes for back-EMF protection.

PIC Microcontrollers & ISP Programming

- Programmed PIC16F84A, PIC18F4550, PIC18F2550; performed In-System Programming (ISP) for firmware updates.

Education

- **Mechatronics Engineering** – Universidad Tecnológica de Tecámac, MX (2022)
Thesis: Automated Control System for Electromechanical Assembly Lines
- **Professional License in Automated Systems & Industrial Computing** – IUT d'Aix-Marseille, France (2020)
Focus: Industrial Networks, Embedded Systems, Automation – MEXPROTEC scholarship

Certifications

- LabVIEW Core 1 & Core 2 (NI Certified)
- Google Cloud: Networking, Security & Infrastructure Foundations
- EF SET English Certificate (B2 Proficient)
- Cisco Networking Academy (Dec 2019)

Interests

- DIY Electronics: Custom PCB design, firmware for IoT devices (ESP32, STM32)
- Emerging Tech: AI integration, blockchain fundamentals
- Outdoor activities: hiking, cycling